

THE ADRIEL SHAMER WILLIS F-007 MODEL

Volume 2: Home-Level CNC Piping Specifications & TEMPEST Emission Mitigation

1. Home-Level CNC Piping Architecture

To implement the F-007 hydroacoustic modulation system within a standard residential framework, the interior fluid channels must be explicitly engineered using localized desktop CNC milling mills. Standard plumbing pipe features uncontrolled internal surface roughness, leading to parasitic turbulent boundary layers that mask high-frequency carrier waveforms (f_1 and f_2).

The model specifies internal fluidic geometry featuring helical acoustic micro-grooves. These structures stabilize Rayleigh-Plateau droplet breakups by locking the boundary fluid into a structured vortex ring before it reaches the nozzle chamber. Machining these paths requires rigid polymer stocks such as Delrin (POM) or PTFE, which provide both chemical corrosion immunity and the precise acoustic dampening required to eliminate macro-mechanical noise.

2. SCADA Signal Control & Radio Frequency EMSEC (TEMPEST)

When high-voltage amplifiers drive the PMN-PT piezoelectric crystals inside a wet plumbing fixture at kilohertz frequencies, the system inherently generates compromising electromagnetic radiation. In security engineering, TEMPEST-level standards dictate that these unintentional emissions must be contained to prevent side-channel eavesdropping or local interception.

To mitigate this risk, the F-007 framework integrates rigorous hardware-level shielding. The entire CNC nozzle assembly functions as a localized Faraday cage. By grounding the metallic outer sleeve of the piping and utilizing double-shielded twisted-pair (S/FTP) cabling directly from the Variable Frequency Drive (VFD), compromising transient signals are entirely attenuated before radiating into the surrounding domestic environment.

3. Material & Shielding Specification Matrix

Component	Material	Engineering Parameter
Internal Core Piping	CNC Machined Delrin (POM)	Helical surface roughness < 1.6 um Ra
Acoustic Isolation Sleeve	High-Density Polyurethane	Dampens micro-frictional pipe whine
TEMPEST Shielding Layer	Woven Copper Conduit Mesh	Attenuates RF emissions up to 10 GHz
Piezo Core Housing	Grade 5 Titanium Base	Withstands cavitation shock stresses

4. Python Hardware Driver Template

The following Python template simulates how the SCADA control node interfaces with the high-voltage amplifier to adjust the carrier waves while filtering transient noise loops:

```
import numpy as np

def compute_tempest_safe_carriers(target_audio_freq, sample_rate=192000):
    """
    Calculates dual carrier frequencies while ensuring the signal
    falls within shielded hardware bandwidth limitations.
    """
    f1_base = 40000.0 # Stable ultrasonic base carrier (40 kHz)
    f2_modulated = f1_base + target_audio_freq

    # Enforce TEMPEST frequency envelope safety guidelines
    if f2_modulated > 60000.0:
        raise ValueError("Frequency exceeds shielded conduit threshold.")

    return f1_base, f2_modulated

# Example execution for a 7 Hz infrasonic difference signal
carrier_1, carrier_2 = compute_tempest_safe_carriers(target_audio_freq=7.0)
print(f"Carrier Channels Initialized: {carrier_1} Hz and {carrier_2} Hz")
```